



Advanced Threat Hunting, SOAR Automation, Incident Response, and Security Operations

Executive Summary

This report summarizes the practical activities performed as part of advanced Security Operations Center (SOC) training. The objective was to develop hands-on experience in threat hunting, security automation, incident response, adversary emulation, evidence analysis, and security metrics reporting.

Various industry-standard security tools, including Wazuh, Elastic Security, TheHive, CrowdSec, Velociraptor, VirusTotal, MITRE Caldera, and Metasploit, were utilized to simulate real-world security operations. Activities focused on threat detection, alert investigation, automated response workflows, forensic analysis, and incident management.

The exercises demonstrated the importance of proactive threat hunting, automation, threat intelligence integration, and structured incident response processes in strengthening organizational security. Overall, the activities provided valuable practical experience and enhanced the skills required for effective SOC operations and cybersecurity incident handling.

Introduction

This assignment focused on developing practical skills in key Security Operations Center (SOC) activities, including threat hunting, SOAR automation, incident response, adversary emulation, evidence analysis, and security reporting. Various security tools were used to simulate real-world cybersecurity operations and investigate security events.

The activities provided hands-on experience in detecting threats, analyzing alerts, automating response actions, and evaluating security controls. Through these exercises, a better understanding of modern SOC workflows and incident handling procedures was achieved.

The knowledge gained from this assignment strengthened both technical and analytical skills required for effective threat detection, investigation, and response within a cybersecurity environment.

1. Threat Hunting Methodologies

Objective

The objective of this exercise was to proactively identify suspicious activities and potential security threats within the environment using threat hunting techniques. The activity focused on analyzing security logs, investigating abnormal system behavior, and detecting indicators of compromise that



may not be identified through traditional alert-based monitoring. Additionally, the exercise aimed to improve understanding of attacker tactics and strengthen threat detection capabilities using security monitoring tools and threat intelligence sources. The exercise also aimed to enhance analytical skills by correlating security events, validating suspicious activities, and improving overall incident detection and investigation capabilities.

Methodology

A hypothesis-driven hunting approach was utilized during this exercise. The investigation began with the assumption that an attacker may attempt to abuse legitimate user credentials to gain elevated privileges and move laterally throughout the environment. Security event logs, authentication records, and privilege assignment events were collected and analyzed using Elastic Security.

Threat intelligence indicators were gathered from AlienVault OTX and compared against observed activity. Event ID 4672 was selected as a primary indicator because it represents special privilege assignments that may be associated with unauthorized administrative activity.

Activities Performed

Multiple search queries were executed within Elastic Security to identify suspicious authentication events, privilege escalation attempts, and unusual account behavior. User activities occurring outside normal working hours were reviewed to determine whether unauthorized access attempts were present. Endpoint telemetry and process execution logs were also analyzed to provide additional context regarding system activities.

Threat intelligence enrichment was performed by comparing observed indicators with publicly available intelligence feeds. Identified findings were mapped to the MITRE ATT&CK framework to classify attacker behavior and improve detection understanding.

Findings

The investigation revealed privileged account activity associated with a test account that received elevated permissions within the monitored environment. While the activity was generated within a controlled laboratory environment, it closely resembled techniques commonly used by attackers after obtaining valid credentials.

The observed activity aligned with MITRE ATT&CK Technique T1078 (Valid Accounts), demonstrating how compromised credentials can be leveraged to gain unauthorized access and perform malicious actions while appearing legitimate.

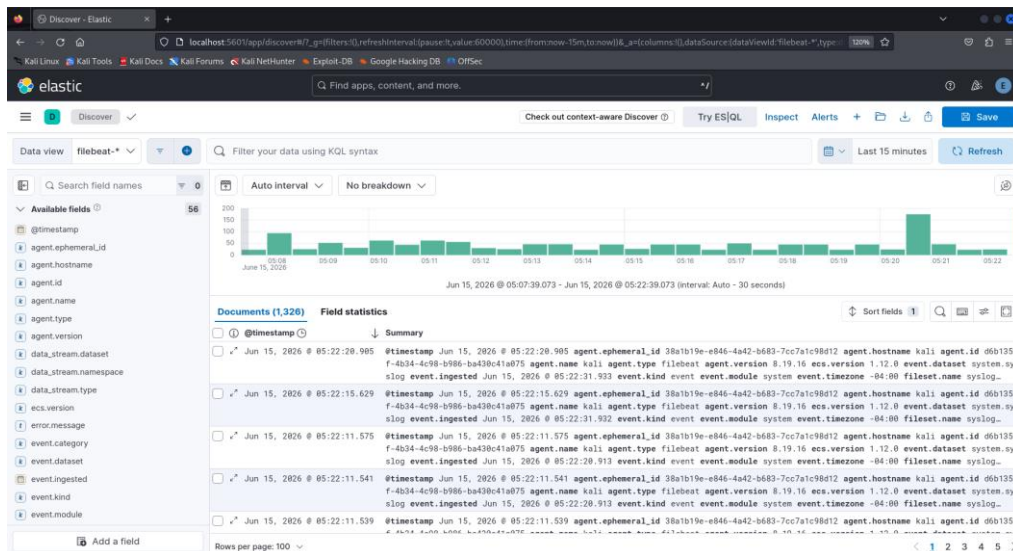


Figure 1: Elastic Discover view displaying collected security events.

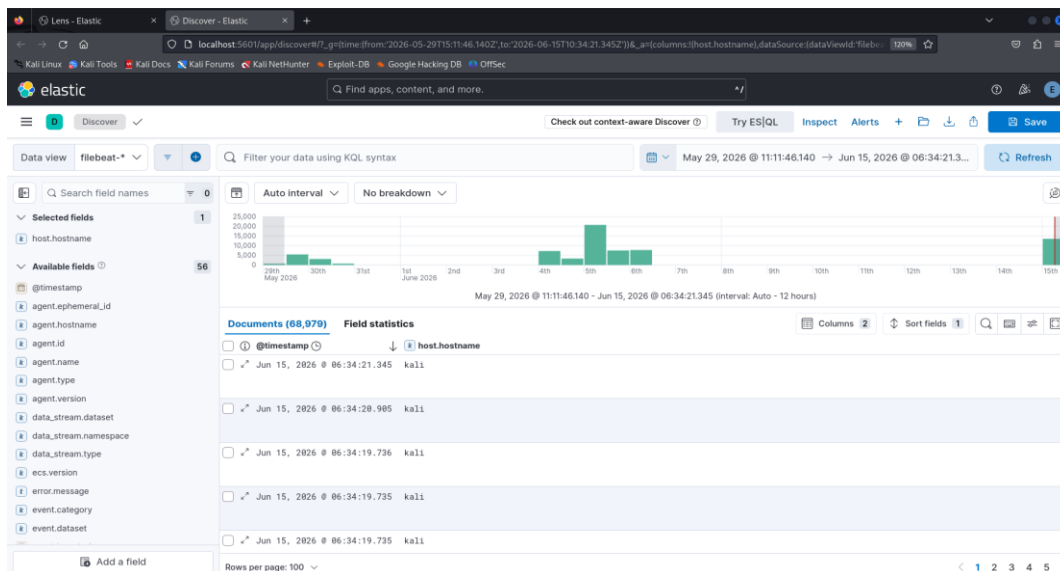


Figure 2: Host activity logs used during threat hunting analysis.

2. Advanced SOAR Automation



Objective

The objective of this exercise was to understand the role of Security Orchestration, Automation, and Response (SOAR) technologies in modern Security Operations Centers. Security teams often face a high volume of alerts and repetitive tasks that consume significant analyst time. SOAR platforms address this challenge by automating routine activities, orchestrating workflows across multiple security tools, and accelerating incident response procedures. Through this exercise, practical experience was gained in designing and evaluating automated response workflows.

Methodology

The exercise focused on developing an automated phishing response playbook. The workflow was designed to automatically validate threat indicators, enrich alerts with threat intelligence, perform containment actions, and create investigation cases for analysts. The objective was to reduce manual intervention while ensuring consistent and repeatable response procedures.

Integration between multiple security tools was simulated to demonstrate orchestration capabilities. Threat intelligence feeds were used to validate indicators, while automated response actions were configured to block malicious IP addresses and generate incident tickets. This workflow represented a typical phishing response scenario encountered within enterprise SOC environments.

Activities Performed

A playbook was designed consisting of multiple automated stages. The workflow began when a phishing-related alert was received from a security monitoring solution. The system automatically performed reputation checks against threat intelligence databases and validated associated indicators of compromise. Once malicious activity was confirmed, the workflow executed predefined containment actions.

Automated containment included blocking malicious IP addresses through CrowdSec and generating incident tickets within TheHive. Notifications were also configured to alert security analysts regarding critical events requiring review. Throughout the process, all actions were documented to maintain visibility and accountability.

The exercise also included reviewing the execution flow to identify opportunities for optimization. Response actions were evaluated to ensure accuracy and effectiveness while minimizing false positives.

Findings

The automated playbook successfully completed all predefined actions and demonstrated the effectiveness of automation within a SOC environment. Threat intelligence enrichment provided valuable context that improved decision-making and reduced investigation time. Automated containment actions prevented potentially malicious connections from continuing while incident tickets ensured proper documentation and follow-up.

The exercise highlighted several advantages of SOAR platforms, including reduced analyst workload, faster response times, improved consistency, and enhanced collaboration between security



technologies. Automation eliminated repetitive manual tasks and enabled analysts to focus on higher-priority investigations.

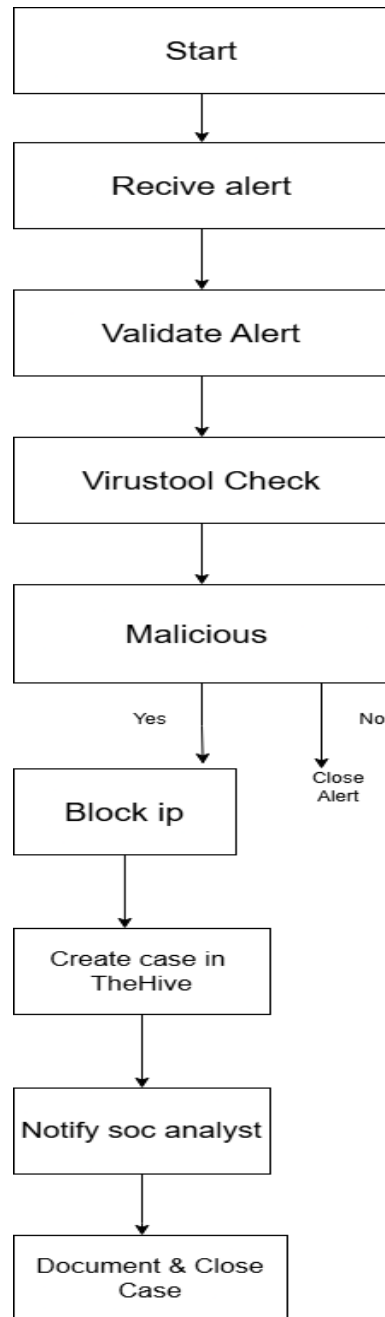


Figure 3: SOAR Playbook Workflow



3. Post-Incident Analysis and Continuous Improvement

Objective

The objective of this exercise was to understand how post-incident analysis contributes to organizational learning and continuous security improvement. Effective incident response does not end with containment and recovery; organizations must also investigate root causes, identify lessons learned, and implement corrective actions to reduce the likelihood of future incidents.

Methodology

A simulated phishing incident was selected for analysis. Root Cause Analysis (RCA) techniques, including the 5 Whys methodology, were used to identify underlying factors that contributed to the incident. Performance metrics such as Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) were reviewed to evaluate response effectiveness.

The analysis focused on understanding how the attack occurred, why existing controls failed to prevent it, and what improvements could be implemented to strengthen defenses. Recommendations were developed based on findings obtained during the investigation process.

Activities Performed

The investigation began by reviewing the incident timeline and documenting all observed events. Analysts examined user actions, email filtering controls, detection mechanisms, and response activities. Each stage of the incident lifecycle was evaluated to determine whether security controls performed as intended.

Using the 5 Whys methodology, analysts traced the incident back to its root causes. The review identified deficiencies in user awareness training, email filtering effectiveness, and detection rule tuning. Additional attention was given to communication procedures and escalation workflows to ensure efficient incident management.

Security metrics were calculated to evaluate performance. MTTD measured the time required to identify malicious activity, while MTTR measured the time required to contain and remediate the incident.

Findings

The analysis revealed that the primary contributing factor was user interaction with a malicious email. Although existing security controls detected the threat, opportunities existed to improve email filtering and user awareness programs. Detection policies also required tuning to improve visibility into suspicious email activity.

Performance metrics indicated acceptable response times; however, improvements in automation and threat intelligence integration could further reduce both detection and response durations. The review also highlighted the importance of conducting regular training sessions to strengthen employee awareness of phishing threats.



4. Alert Triage and Threat Validation

Objective

The objective of this exercise was to develop practical experience in analyzing, prioritizing, and validating security alerts. Security monitoring systems often generate large volumes of alerts, many of which may be false positives. Effective triage processes help analysts identify genuine threats and allocate resources appropriately.

Methodology

The alert triage process began by reviewing security alerts generated by Wazuh and categorizing them based on severity, source, and potential impact. Each alert was analyzed to identify relevant indicators of compromise, including suspicious IP addresses, file hashes, and user activities. Threat intelligence sources such as VirusTotal were used to validate indicators and determine whether the observed activity was associated with known malicious behavior.

Validated alerts were prioritized according to risk level and escalated for further investigation when necessary. Additional context was gathered through log analysis and correlation of related events to support accurate incident classification and response decisions.

Activities Performed

Security alerts generated by Wazuh were reviewed and categorized according to severity and potential impact. Suspicious file download events were selected for detailed investigation. Threat intelligence enrichment was performed using VirusTotal to determine whether associated file hashes had been identified as malicious.

Alert context was analyzed to identify affected systems, source IP addresses, user accounts, and associated indicators of compromise. High-risk alerts were escalated for additional investigation while low-risk alerts were documented and monitored.

Findings

The investigation confirmed malicious characteristics associated with the analyzed file. Threat intelligence validation provided additional context regarding malware classifications and known indicators. Based on the findings, the alert was escalated for containment and remediation actions.

The exercise demonstrated how automated threat intelligence enrichment improves investigation efficiency by providing analysts with immediate access to relevant information. This reduces the time required to validate threats and improves prioritization accuracy.

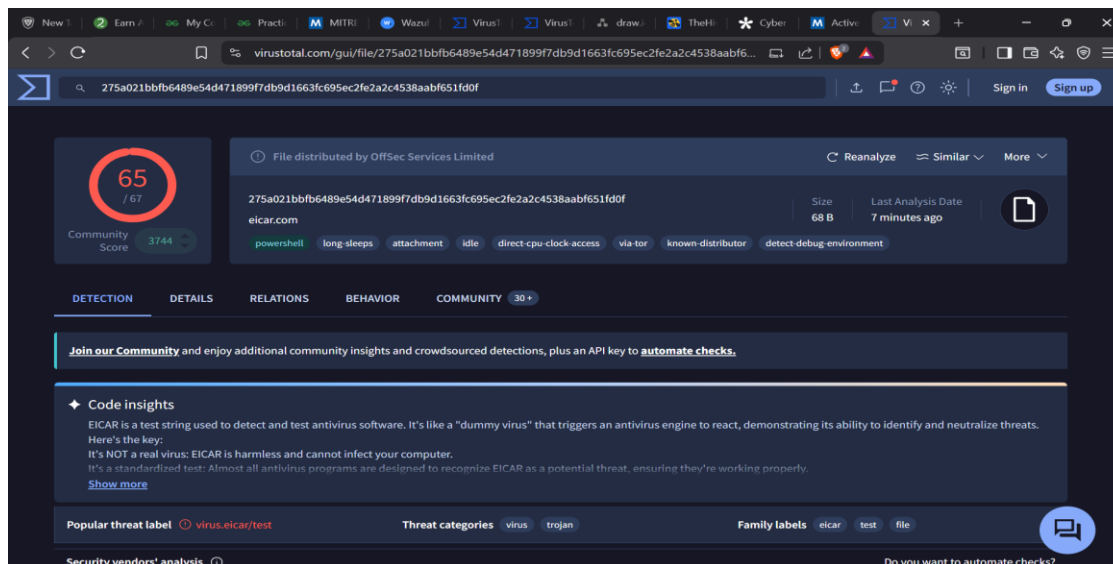


Figure 4: VirusTotal Detection Summary

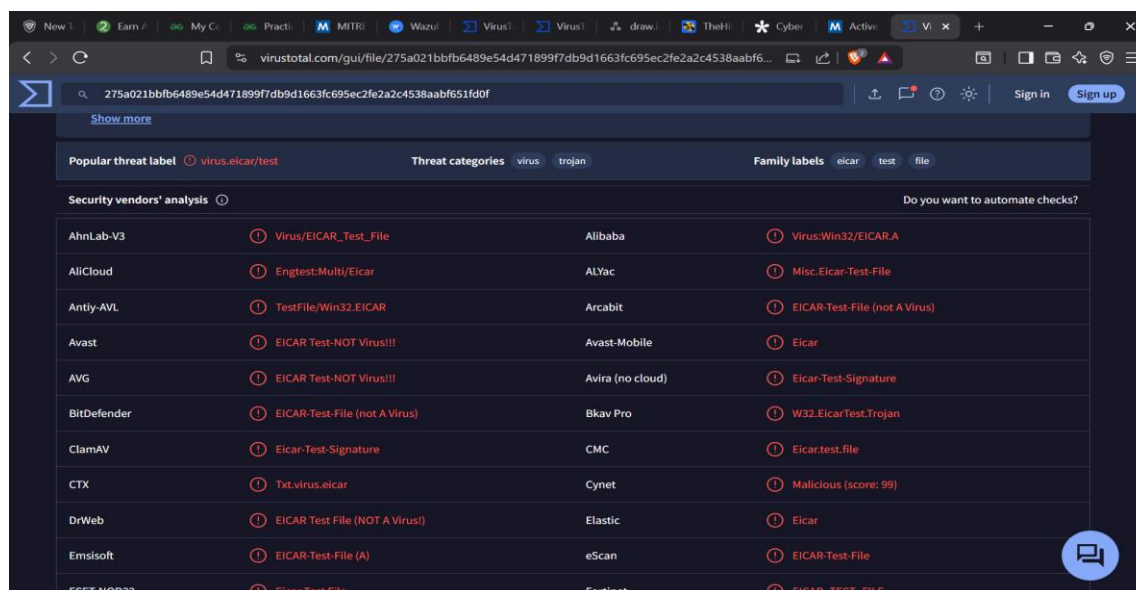


Figure 5: VirusTotal Vendor Analysis



5. Evidence Analysis

Objective

The objective of this exercise was to understand the processes involved in collecting, preserving, and analyzing digital evidence during cybersecurity investigations. Proper evidence handling is essential for maintaining data integrity and supporting investigative conclusions.

Methodology

The evidence analysis process involved collecting and examining digital artifacts from endpoint and network sources using forensic tools such as Velociraptor and FTK Imager. Relevant system logs, network connections, and process execution records were reviewed to identify suspicious activities associated with the investigation.

All collected evidence was documented and preserved following chain-of-custody procedures to maintain integrity throughout the analysis process. Hash verification was performed to ensure evidence remained unaltered, and findings were correlated with incident data to support investigative conclusions and response activities.

Activities Performed

Evidence was collected from both endpoint and network sources using Velociraptor and FTK Imager. Analysts reviewed active network connections, process execution records, and system activity logs to identify suspicious behavior. All evidence collection activities were documented according to established forensic procedures.

Chain-of-custody documentation was maintained throughout the investigation. Cryptographic hash values were generated to verify evidence integrity and ensure that collected artifacts remained unchanged during analysis.

Findings

Analysis revealed several unusual outbound network connections that warranted further investigation. While the activity occurred within a controlled training environment, it demonstrated how forensic evidence can provide critical insights into attacker behavior and system compromise.

The exercise also highlighted the importance of preserving evidence integrity. Proper documentation and verification procedures ensure that findings remain reliable and defensible throughout the investigative process.

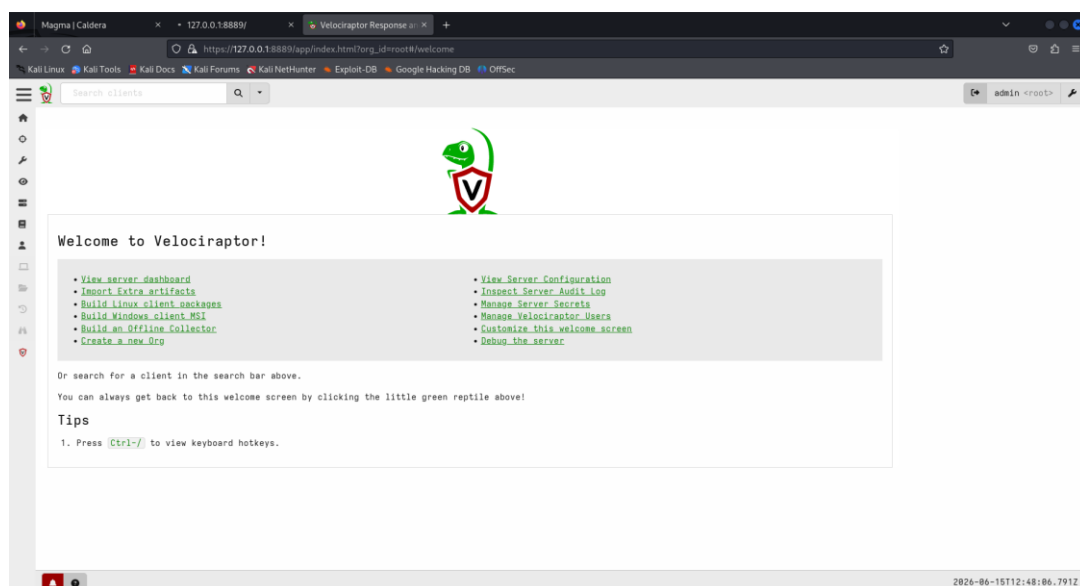


Figure 6: Velociraptor Dashboard

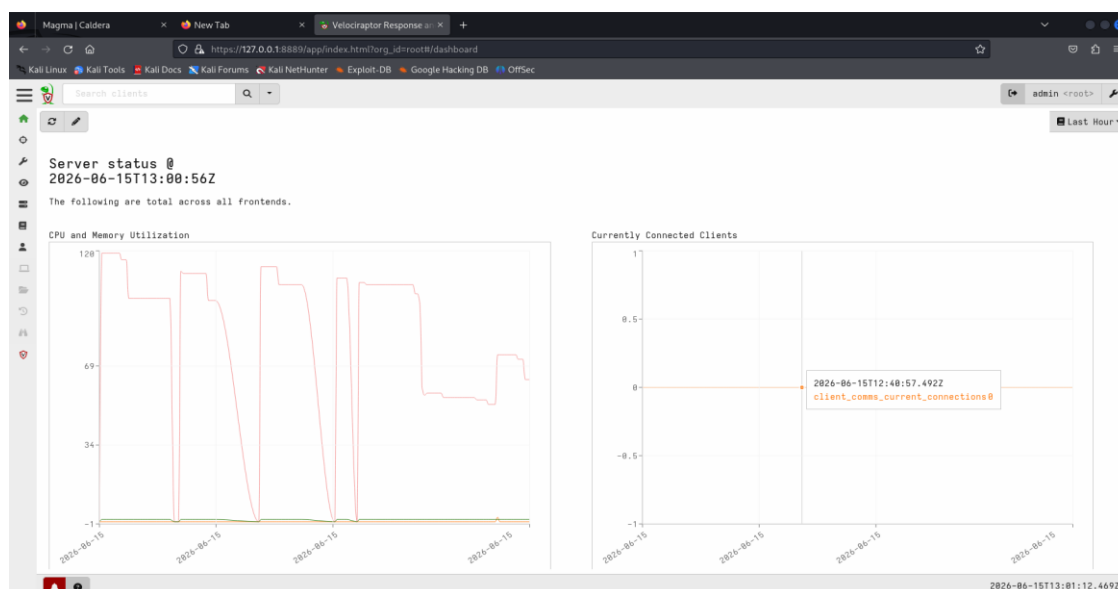




Figure 7: Velociraptor Monitoring Dashboard

6. Adversary Emulation Techniques

Objective

The objective of this exercise was to simulate real-world attacker behavior and evaluate the effectiveness of existing security monitoring and detection capabilities. Adversary emulation helps organizations identify gaps in security controls by replicating techniques commonly used by threat actors.

Methodology

MITRE Caldera was used to perform adversary emulation activities based on the MITRE ATT&CK framework. A test operation was created to simulate attacker actions within a controlled environment. The generated activities were monitored through security tools to determine whether the simulated attacks could be detected and investigated successfully.

Activities Performed

A Caldera operation was configured and executed to emulate common attack techniques. The platform utilized predefined adversary profiles and abilities to simulate attacker behavior. During execution, security logs and events were generated, allowing monitoring tools to capture and analyze the activity.

The operation results were reviewed to verify whether security controls could detect the simulated techniques. Generated logs were correlated with monitoring data to understand attack visibility and detection coverage.

Findings

The simulated adversary activities were successfully executed through MITRE Caldera. Security monitoring tools generated logs related to the operation, demonstrating visibility into attacker-like behavior within the environment. The exercise confirmed that monitoring solutions were capable of capturing relevant events associated with the simulated attack techniques.

The activity also highlighted the importance of continuous validation of detection rules and security controls. Adversary emulation provides valuable insight into how attackers may operate and helps organizations strengthen their defensive capabilities.

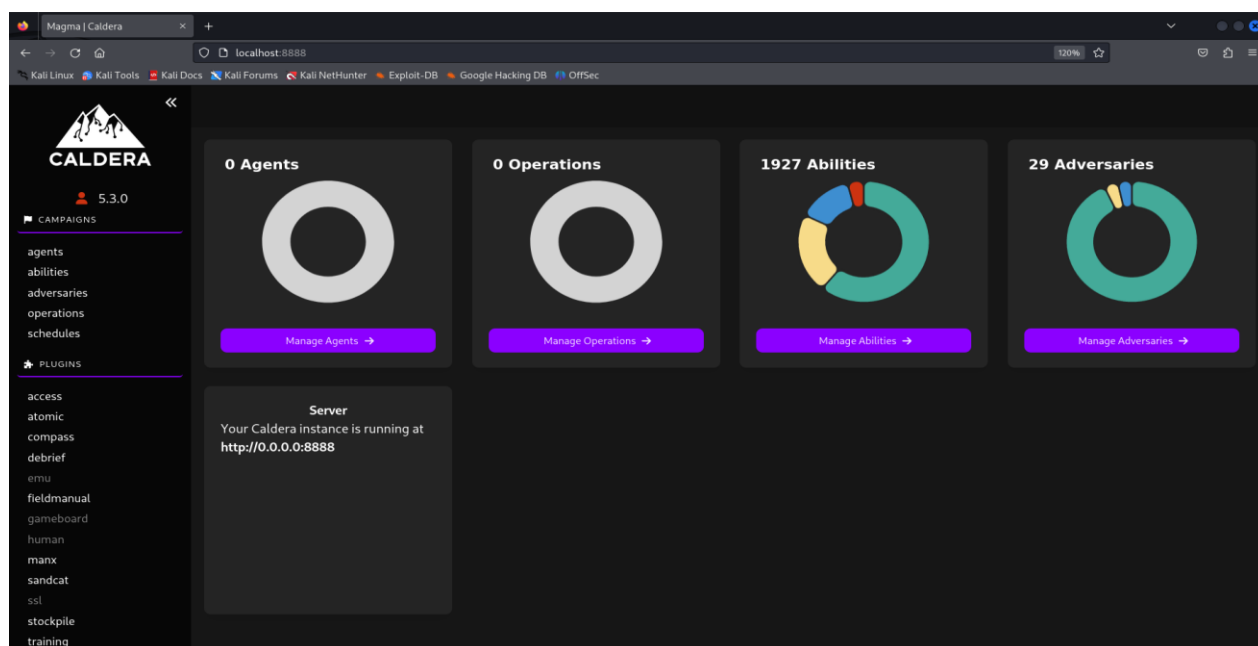


Figure 8: MITRE Caldera Dashboard

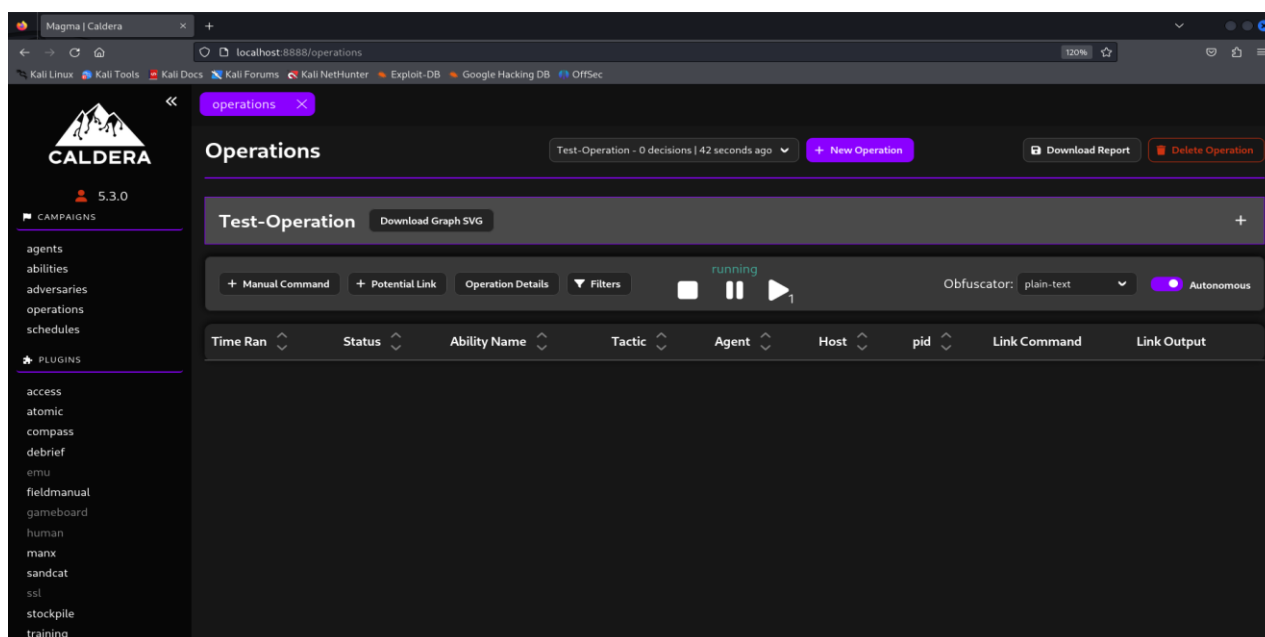


Figure 9: MITRE Caldera Test Operation Execution



Challenges Faced

During the completion of this assignment, several challenges were encountered while configuring and integrating multiple security tools. Understanding the workflows between Wazuh, TheHive, CrowdSec, Elastic Security, and MITRE Caldera required additional research and troubleshooting to ensure proper functionality.

Threat hunting and alert analysis required careful interpretation of logs and security events to distinguish legitimate activities from potentially malicious behavior. Correlating alerts with threat intelligence data and validating indicators of compromise also required attention to detail and analytical thinking.

Another challenge involved understanding incident response workflows and automation processes. Designing response playbooks, documenting findings, and maintaining proper evidence handling procedures required a structured approach to ensure accuracy and consistency.

Despite these challenges, the activities provided valuable hands-on experience and improved understanding of SOC operations, threat detection, incident response, and security monitoring practice.

Tools Used

Tool	Purpose
Wazuh	Security Monitoring & Alerting
Elastic Security	Log Analysis & Threat Hunting
VirusTotal	Threat Intelligence Validation
TheHive	Incident Management
Velociraptor	Digital Forensics & Evidence Collection
MITRE Caldera	Adversary Emulation
CrowdSec	Automated Blocking & Response

Conclusion

This report provided practical experience in key areas of Security Operations Center (SOC) activities, including threat hunting, SOAR automation, incident response, evidence analysis, adversary emulation, and security reporting. Through the use of industry-standard tools such as Wazuh, Elastic Security, TheHive, CrowdSec, Velociraptor, and MITRE Caldera, valuable insights were gained into modern cybersecurity operations.

The exercises demonstrated the importance of proactive threat detection, automated response workflows, effective incident management, and continuous improvement processes. The capstone



project successfully combined all learned concepts into a complete incident response lifecycle, from detection and investigation to containment and reporting.

Overall, the activities strengthened both technical and analytical cybersecurity skills while providing hands-on experience with real-world SOC workflows. The knowledge gained will support future responsibilities in security monitoring, threat detection, incident analysis, and response operations.

References

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